

Class 26 Math Induction

Goal

Learn to use the technique of mathematical induction.

Induction proofs are typically used to prove statements that depend on integers.

Examples

There are 2^n subsets of a set of size n .
 $\forall n \geq 0$

The sum of the first n positive odd integers is n^2 $\forall n \geq 1$

$$D_x [x^n] = nx^{n-1} \quad \forall n \geq 1$$

Weak Principle of Math Induction

Let $p(n)$ be a statement defined for $n \in \mathbb{Z}$ and let $a \in \mathbb{Z}$. If $p(a)$ is true and if $p(k) \Rightarrow p(k+1)$ for all $k \geq a$ then $p(n)$ is true for all $n \geq a$.

Example 1: Prove $\sum_{i=1}^n 2i = n^2 + n \forall n \geq 1$ This is $p(n)$

Base Case : If $n=1$.

$$\sum_{i=1}^1 2i = 2 = 1^2 + 1$$

So the statement $p(1)$ is true.

Weak induction step

[show if $p(k)$ is true, then $p(k+1)$ is true]

Assume the statement is true when $n=k$

Let k be any integer ≥ 1 . Prove

$$\text{if } \sum_{i=1}^k 2i = k^2 + k \text{ then } \sum_{i=1}^{k+1} 2i = (k+1)^2 + (k+1)$$

Given $\sum_{i=1}^k 2i = k^2 + k$ we know

$$\sum_{i=1}^{k+1} 2i = \underbrace{2+4+6+\dots+2k}_{k^2+k} + 2(k+1) = k^2 + k + 2k + 2$$

$$= (k^2 + 2k + 1) + (k+1) = (k+1)^2 + k+1$$

Conclusion: Since the statement is true for $n=1$ and is true for $n=k+1$ whenever true for $n=k \geq 1$ we know the statement is true for all integers $n \geq 1$

Example 2: Prove $2^n < (n+2)! \forall n \geq 0$

Base Case: If $n=0$ $2^0=1$ and $(0+2)!=2$
So $2^0 < (0+2)!$

Let k be any integer ≥ 0 .

Prove: If $2^k < (k+2)!$ then $2^{k+1} < \overbrace{(k+1)+2}^{(k+3)}!$

Given $2^k < (k+2)!$ and $k \geq 0$

$$2^{k+1} = 2 \cdot 2^k < 2(k+2)! < (k+3)(k+2)! = (k+3)!$$

So the statement is true for $n=k+1$
whenever true for $n=k$.

Conclusion: Since the statement is true for $n=0$ and is true for $n=k+1$ whenever true for $n=k \geq 0$, we know the statement is true for all integers $n \geq 0$.